
FLUOR DANIEL

MASONRY BUILDING DESIGN**PURPOSE**

This practice establishes guidelines for the design of hollow core masonry structures.

SCOPE

This practice consists of a narrative, attachments and one sample building design. Discussions focus on loading, materials, and basic design criteria.

This practice includes the following major sections:

- DESIGN CONDITIONS
- ROOF AND FLOOR FRAMING
- MATERIALS
- ALLOWABLE STRESSES
- CONNECTIONS
- WALL DESIGN
- REFERENCES
- ATTACHMENTS

APPLICATION

This practice may be used for the design of CMU office and control buildings typically found in industrial plants or refineries. This practice does not apply to buildings designed for blast resistant criteria.

**DESIGN
CONDITIONS**

Masonry structures must be designed to resist a wide variety of loading conditions, both vertical and horizontal. The masonry elements may be designed as bearing or nonbearing, shearwalls or nonshearwalls. For each of these cases, proper loading conditions must be considered, and proper detailing must be done to ensure that elements perform as they are intended. As an example, walls that are intended to be strictly partition walls must be detailed to eliminate possible bearing and diaphragm load transfer. Conversely, walls designed as bearing/shearwalls must be detailed to properly develop all design loads.

Design Criteria

Design loads should follow requirements of the applicable code of the governing jurisdiction, or of project specific criteria. As an example, sample references to the UBC (1994 Uniform Building Code) have been used in this practice.

Vertical Loads

Vertical loads consist of dead loads in conjunction with some or all of the other loads listed below. These loads result in both an axial component and a flexural component. Flexural stresses are caused by eccentricity of the floor connection as well as P-delta effects.

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Dead

Dead loads may include the following:

- Diaphragm (plywood or decking)
- Subframing
- Beams and girders
- Mechanical and electrical equipment
- Roofing materials
- Ceiling assemblies (if any)
- Roof mounted equipment
- Wall weight

Live

Live loads are as specified in the applicable codes (UBC Table 16A) or project specific criteria.

Snow

Snow loads will be designed as required by local criteria.

Wind

Vertical effects of wind loads should be considered, particularly for lightweight roof systems where suction may create net upward loads.

Load Combinations

Loads will be combined as specified in the applicable codes or by project criteria.

Horizontal Loads

Horizontal loads generally include some or all of the following:

- Seismic
- Wind
- Snow drift
- Active earth pressure
- Blast wind (not covered)
- Internal explosion containment (not covered)

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Out-Of-Plane Loads

Out-of-plane loads are those which act perpendicular to the plane of the wall. These loads are distributed to the horizontal floor elements through beam action of the wall. These forces may be caused by the following:

- Seismic
- Wind
- Snow drift
- Lateral earth pressure

In-Plane Loads

In-plane loads are distributed by the floor diaphragms to the shearwalls. These forces may be caused by:

- Seismic
- Wind

**ROOF AND FLOOR
FRAMING**

Floor and roof framing can be designed using a wide variety of construction materials and systems. The type of construction will depend upon many factors including fire rating, local availability, cost, loading, durability, or even client preference.

Framing systems can be designed as total vertical load carrying systems (Building Frame Systems as defined in the UBC) taking all vertical loads, or by utilizing interior and exterior bearing walls (Bearing Wall Systems). When used as shearwalls, these two systems are assigned the seismic coefficient R_w equal to 8 and 6 respectively.

Types Of Framing

Many types of construction materials can be utilized. During the schematic phase, all viable materials and schemes should be considered to optimize cost and performance.

Wood Framing

Many types of wood framing systems are utilized for floor and roof framing. Sawn lumber, plywood web joists, glulam beams, and open web wood joist can provide very economical framing solutions.

Plywood is generally used to carry vertical loads between joists, and to act as the horizontal diaphragm. Lateral analysis is performed treating the plywood as a flexible diaphragm, distributing loads by tributary area or statics. Walls must be positively anchored to the diaphragm (UBC Section 1611) for a minimum of 200 pounds per lineal foot. Effects on subdiaphragms to resist these anchor forces must also be considered. (Other considerations are covered in UBC Section 2316.)

Diaphragm deflection must be checked to control wall cracking.

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Steel Framing

Steel joists and beams, in combination with metal decking, can provide strong, durable, and economic framing solutions. Depending on the design loads and conditions, different combinations of these systems may offer better performance.

Roof systems are generally designed with 1- 1/2 inch metal deck profiles without concrete fill. Insulating concrete fill may be used, but does not provide any structural support. This type of deck is preferred when a minimal amount of roof equipment is present, or when heavier equipment is supported with independent framing above or below the roof. These decks provide moderate diaphragm performance, and generally act as semiflexible to flexible diaphragms, depending upon deck rigidity and beam spacing. Diaphragm capacity is controlled through spacing of welds or screws at supports.

Floor systems utilize deck profiles of 2 to 3 inches in depth. Due to the loads and function, these decks are filled with concrete to provide a level floor surface. These decks are often deformed, which results in a composite floor slab. Sometimes, however, thin profile nondeformed roof deck is used simply as a form for casting thicker slabs. In this case, no composite action is used. The resulting slabs are generally treated as rigid diaphragms. In such cases, wall rigidities, center of gravity, and torsion must be taken into account in a rigid diaphragm analysis.

Steel framing may be designed to act compositely with the concrete filled decks. In such designs, headed studs are used to develop the horizontal shear at the interface of the steel and concrete. Additional detailing and construction costs must be weighed against the steel savings to determine whether there is an advantage to designing composite beams. The use of lightweight concrete can also be considered to further reduce beam weight, but the benefits must be weighed against higher unit concrete prices.

Concrete Framing

The use of cast-in-place or precast concrete comes into play for heavily loaded or specialty design structures. These systems provide strong, durable structures which can be adapted to food plants, biotech facilities, blast resistant, and blast containment structures.

Cast-in-place systems require field forming and tend to lengthen construction time, but these systems allow for more flexibility in field changes. Most systems are 1-way slabs, with or without intermediate beam supports. Concrete slabs are analyzed as rigid diaphragms.

Precast systems are often more economical, and can be erected very quickly. These systems, however, require thorough detailing to ensure constructibility and proper performance. Precast systems, such as prestressed planks, are generally set in place and designed only for vertical loads. They rely on topping slabs to act as a diaphragm to transfer lateral loads to independent lateral resisting systems such as shearwalls.

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MASONRY BUILDING DESIGN**MATERIALS**

Standards for material quality are specified in the applicable code.

**Concrete Masonry
Units**

CMU (Concrete masonry units) are specified as Grade N (exposed to weather or moisture penetration), or Grade S (not exposed to weather). Either grade of block may be specified as Type I (moisture controlled), or Type II (nonmoisture controlled) depending upon location, use, and construction methods. The Type I and II designations are used primarily to control shrinkage due to moisture loss.

Mortar

Mortar is one of the basic components of masonry construction, serving many purposes:

- Serves as a bedding material for placing and leveling the block unit.
- Serves to bond the units together.
- Provides shear strength and compressive strength between units.
- Allows for movement and elasticity between units.
- Seals irregularities between units and provides a weather tight seal.
- Can be used to accent architectural appearance by utilizing various joint tools.

Mortar type is specified dependent upon the class of construction. Types of mortar are as follows:

Type	Suitability of Construction
M	Subjected to high compressive loads, severe frost action, lateral earth pressure, high wind, or seismic loads.
S	Structures requiring high bond strength, but subject only to normal compressive loads.
N	General above grade construction.
O	Nonloadbearing walls.

For most design structures, Type M or S should be specified, with Type M being the most preferred.

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Grout

Grout is a fluid concrete mix which serves to bond the reinforcing steel and masonry units together to act as a composite section. The relatively high water to cement ratio is compensated by the absorbent masonry units, and allows for thorough filling of cells.

Reinforcing

Reinforcing steel generally follows ASTM (American Society for Testing and Materials) A615, Grade 40 to 60. Deformed bars are placed in solid grouted cells to add flexural strength to the section.

ALLOWABLE STRESSES

Allowable stresses for hollow concrete masonry units are addressed in the applicable codes. These sections should be thoroughly reviewed by the Design Engineer.

Specified Compressive Strength

As an example, the specified design compressive strength, f_m , will be based on 1 of 3 procedures (UBC criteria).

1. Prism tests as specified in contract documents for materials and practices used for the project.
2. If accepted prism tests are on file, f_m may be selected on a value not exceeding 75 percent of the average record values.
3. From Table 21-D for the appropriate mortar type and strength.

Method 3 will reduce testing time and costs, but results in a design strength from 38 percent to 75 percent of the actual ultimate strength of the units. The assumed design strength, f_m , should be clearly specified in the contract documents. In areas designated as Seismic Zone Numbers 3 and 4, the f_m from Table 21-D will not exceed 1,500 psi.

Special Inspection

Section 1701 of the UBC requires special inspection for all Structural Masonry operations. This inspection may be omitted when the allowable design masonry stresses are reduced by 1/2. It is recommended, however, that walls used as major structural elements, particularly those in Seismic Zones Numbers 3 and 4, utilize special inspection.

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CONNECTIONS

Connections between floor levels should be designed to develop both vertical and horizontal loads as obtained from analysis, or for code minimum criteria.

Bolts

Allowable shear and tension values for embedded anchor bolts in masonry are discussed in UBC Section 2107. Allowable shear values are shown in Table No. 21-F for various compressive strengths; whereas, allowable tension values are found as the lesser value of Table No. 21-E-1 or 21-E-2.

Allowable bolt values are governed by the following equations:

Tension

The allowable tension load, B_t , of anchor bolts is governed by the lesser of

$$B_t = 0.5A_p\sqrt{f'_m}$$

$$B_t = 0.2 A_b f$$

where

A_p is the lesser of

$$A_b = \pi l_b^2$$

$$A_b = \pi l_{be}^2$$

and where

l_b = Embedment length of bolt

l_{be} = Edge distance of bolt

when the projected areas of adjacent anchor bolts overlap, A_p of each bolt will be reduced by 1/2 the overlapping area.

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Shear

The allowable shear load, B_v , on anchor bolts is governed by the lesser of:

$$B_v = 350^4 \sqrt{f'_m A_b}$$

$$B_v = 12 A_b f$$

where

A_b is the same as described above.

When edge distances l_{be} are less than 12 bolt diameters, the value of B_v will be reduced by linear interpolation to zero at an l_{be} of 1- 1/2 inches.

$$B'_v = B_v \times \frac{12 \text{ bolt diameters} - l_{be} - 1 - 1/2 \text{ inches}}{12 \text{ bolt diameters}}$$

Spacings

Anchor bolts will have a minimum edge distance l_{be} of 1- 1/2 inches, and a minimum embedment of 4 bolt diameters. Bolts will have a minimum spacing of 12 bolt diameters.

Bearing

When loads are transferred through bearing, the allowable stresses will be governed by UBC Section 2107.3.8, where allowable bearing is determined by

$$F_{br} = 0.26 f'_m \quad (\text{on full area})$$

$$F_{br} = 0.38 f'_m \quad (\text{on } 1/3 \text{ area or less})$$

WALL DESIGN

Wall elements are designed as bearing walls or nonbearing walls, shearwalls on nonshearwalls, depending upon how they are detailed. The design of these elements is dependent on the direction to the lateral load. For walls perpendicular to the lateral load, elements are designed as beam columns resisting vertical loads as well as out of plane loads. For walls parallel to the lateral load, elements are designed as shearwalls resisting in plane shear loads.

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Distribution Of Loads

Vertical

Walls supporting concentrated loads must be constructed in a running bond. The effective width of the compression element will not exceed the center to center distance between loads, nor the width of the bearing area plus 4 times the wall thickness. The effective area being the lesser of:

$$\begin{aligned} A_{eff} &= (l) * (\text{center to center spacing}) \\ &= (l) * (\text{bearing length} + 4t) \end{aligned}$$

Horizontal (Out-Of-Plane)

The effective width of flexural members resisting out of plane loads is dependent upon the type of construction. For running bonds, the effective width b will not exceed the spacing of the reinforcing bars, nor 6 times the wall thickness. For stack bonds, the effective width b will not exceed the spacing of the reinforcing bars, nor 3 times the wall thickness.

Walls can be designed to span either horizontally or vertically, depending upon support locations and resulting stiffnesses.

Horizontal (In-Plane)

In-plane shear forces which are resisted by masonry shearwalls will be designed for 1.5 times the calculated wind or seismic load from analysis. Loads are distributed throughout the wall proportional to the relative rigidities of the individual piers. Rigidities and deflections are evaluated using the following guidelines:

$$R = \frac{1}{\Delta}$$

■ Fixed-fixed piers:

$$\Delta F = 0.1 \left(\frac{h}{d} \right)^3 + 0.3 \left(\frac{h}{d} \right)$$

■ Cantilevered piers:

$$\Delta C = 0.4 \left(\frac{h}{d} \right)^3 + 0.3 \left(\frac{h}{d} \right)$$

The first part of each equation accounts for the flexural deformation of the pier, the second part for the shear deformation.

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Rigidities and deflections for combinations are evaluated using the following equations:

$$\Sigma R = R_1 + R_2 + \dots + R_n$$

$$\Delta = \frac{1}{\Sigma R}$$

- Walls in series

$$\Sigma \Delta = \Delta_1 + \Delta_2 + \dots + \Delta$$

$$R = \frac{1}{\Sigma \Delta}$$

Wall Elements

Wall elements should follow these basic guidelines:

Dimensions

- Minimum thickness = 6 inches
- Span to thickness ratio

$$h'/t < 30$$

$h'/t > 30$ follow slender wall criteria (Section 2107 of the UBC)

where

$$h' = h \text{ for pin pin}$$

$$h' = 2h \text{ for cantilever}$$

$$h' < h \text{ when substantiated}$$

and

$$h = \text{Unsupported span (height or length)}$$

$$h' = \text{Effective span}$$

$$t = \text{Effective wall thickness}$$

Reinforcing

- $(A_{sh})_{min} = (A_{sv})_{min} \geq .0007 A_g$
- $A_{sh} + A_{sv} \geq .002 A_g$
- Spacing $\leq 4' - 0''$
- No. 3 bar minimum

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Stresses

- Axial:

$$f_a = P/A_e$$

where

P = Design axial load

A_e is the smaller of:

$$A_e = t \text{ (bearing length + } 4t)$$

$$A_e = t \text{ (spacing of loads)}$$

f_a = axial stress in masonry

$$F_a = 0.20 f'_m \left[1 - \left(\frac{h'}{42t} \right)^3 \right]$$

F_a = Allowable axial compressive stress

t = Effective wall thickness

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■ Flexure:

Masonry

Steel

$$f_b = \frac{M}{bd^2} \times \frac{2}{jk}$$

$$f_s = \frac{M}{A_s jd}$$

$$F_b = 0.33 f'_m$$

$$F_s = 0.5 F_y$$

$$= 2,000 \text{ psi max}$$

$$= 24,000 \text{ psi max}$$

where

f_b = Bending stress in masonry

F_b = Allowable bending stress in masonry

f_s = Steel stress

F_s = Allowable steel stress

M = Applied moment

b = Effective width of compression face

d = Distance from extreme compression face to center of tensile reinforcement

A_s = Tensile steel area

k = Ratio of compression depth in masonry to d

$$k = \sqrt{2np + (np)^2} - n$$

$$j = \left(1 - \frac{k}{3}\right)$$

$$n = \frac{E_s}{E_m}$$

$$p = \frac{A_s}{bd}$$

E_s = Modulus of elasticity of steel

E_m = Modulus of elasticity of masonry

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■ Shear

$$f_v = \frac{V}{b_j d}$$

$$F_v = 1.0 (f'_m)^{\frac{1}{2}}$$

$$= 50 \text{ psi ma}$$

where

V = Applied shear

f_v = Shear stress in masonry

F_v = Allowable shear stress in masonry

Bearing

$$f_{br} = \frac{P}{A_{br}}$$

$$F_{br} = 0.26 f'_m \quad (\text{on full area})$$

$$F_{br} = 0.38 f'_m \quad (1/2 \text{ area or less})$$

where

f_{br} = Bearing stress

F_{br} = Allowable bearing stress

P = Applied load

A_{br} = Bearing area

Combined

Check combined axial and flexure by the following equation:

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} \leq 1.00 \quad (\text{or } 1.33 \text{ for seismic or wind})$$

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Shearwalls

$$f_s = \frac{v}{jbd}$$

for masonry taking shear:

$$\begin{aligned} \frac{M}{Vd} < F_v &= \frac{1}{3} \left[4 - \left(\frac{M}{Vd} \right) \right] (f'_m)^{\frac{1}{2}} \\ &= \left(80 - 45 \frac{M}{Vd} \right) \text{ psi ma} \end{aligned}$$

$$\begin{aligned} \frac{M}{Vd} \geq F_v &= 1.0 (f'_m)^{\frac{1}{2}} \\ &= 35 \text{ psi ma} \end{aligned}$$

for rebar taking shear:

$$\begin{aligned} \frac{M}{Vd} < F_v &= \frac{1}{2} \left[4 - \left(\frac{M}{Vd} \right) \right] (f'_m)^{\frac{1}{2}} \\ &= \left(120 - 45 \frac{M}{Vd} \right) \text{ psi ma} \end{aligned}$$

$$\begin{aligned} \frac{M}{Vd} \geq F_v &= 1.5 (f'_m)^{\frac{1}{2}} \\ &= 75 \text{ psi ma} \end{aligned}$$

rebar spacing S is governed by:

$$S = \frac{A_v F_s d}{V}$$

where

A_v = Area of steel required for shear reinforcement

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Bond Beams

Bond beams are continuous reinforced elements in the wall designed to transfer horizontal loads from the diaphragm into the shear wall. These loads are a result of either chord forces (in which the bond beam acts as a flange), or drag forces (in which the bond beam acts as a collector). The bond beam reinforcing is designed to resist tension, while the grouted masonry is designed to resist compression.

Lintel Beams

Lintel beams are used to transfer for vertical loads over openings in the wall such as doors and windows. These loads may be a result of floor and roof framing, live loads, and other dead loads including weights from walls above. Additionally, external walls should be checked for effects of out of plane wind loads.

Columns And Pilasters

Columns and pilasters are designed to meet the stress limitations outlined previously. Axial forces are assumed to be applied through the centroid with uniform distribution over the effective area.

$$f_a = \frac{P}{A_c}$$

$$F_a = \frac{P_a}{A_c}$$

$$P_a = (0.20 f'_m A_c + 0.65 A_s F_{sc}) \left[1 - \left(\frac{h'}{42t} \right)^3 \right]$$

where

f_a , P , A_c , P_a , f'_m , F_a , A_s , t and h' are previously defined.

F_{sc} = Allowable compressive stress in column reinforcement

Bending stresses are designed using the same criteria as before. P-delta effects of eccentric bends are to be added as an additional bending moment.

The area of vertical (longitudinal) reinforcement will not be less than $0.005 A_c$, nor greater than $0.04 A_c$. A minimum of 4 No. 3 bars shall be provided.

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All longitudinal bars must be enclosed by a lateral tie, placed no less than 1- 1/2 inches or greater than 5 inches, from the face of the column. Ties spacing is governed by the following:

- $\leq 16 d_b$ of longitudinal bars
- $\leq 48 d_b$ of ties
- ≤ 18 inches
- $\leq$ least dimension of the column

where

d_b = Bar diameter

A minimum size of No. 3 bars for longitudinal ties is recommended.

REFERENCES

ASTM (American Society for Testing and Materials) A615

UBC (Uniform Building Code, 1994 Edition)

Structural Engineering

Practice 670.215.1215: Wind Load Calculations

Structural Engineering

Practice 670.215.1216: Earthquake Engineering

ATTACHMENTS

Attachment 01:

Typical Interior Bearing Wall Footing

Attachment 02:

Typical Wood Ledger Details

Figure 1. Wood Ledger At Purlins

Figure 2. Wood Ledger At Sub-Purlins Or Rafters

Attachment 03:

Typical Steel Ledger Angle Details

Figure 1. Ledger Elevation At Purlins

Figure 2. Ledger Elevation At Sub-Purlins

Figure 3. Ledger At Change In Roof Slope

Figure 4. Splice At Ledger Angle

Attachment 04:

Figure 1. Typical Wood Nailer

Figure 2. Typical Bearing Plate

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Attachment 05:

Typical Wood Framing Details

- Figure 1. Wood Framing Perpendicular To Single Story Wall #1
- Figure 2. Wood Framing Perpendicular To Single Story Wall #2
- Figure 3. Wood Framing Parallel To Single Story Wall

Attachment 06:

- Figure 1. GLB Beam Seat
- Figure 2. Steel Beam Seat

Attachment 07:

Steel Joist Seat Detail

Attachment 08:

Steel Joist To CMU Wall Detail

Attachment 09:

Sample Design 1: Masonry Building

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TYPICAL WOOD LEDGER DETAILS

Figure 1: Wood Ledger at Purlins

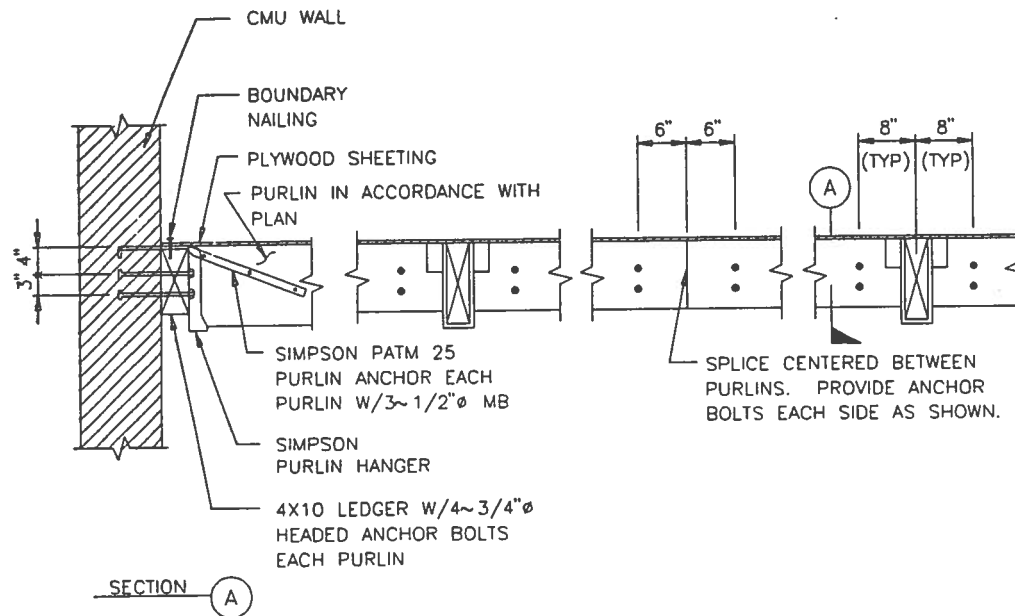
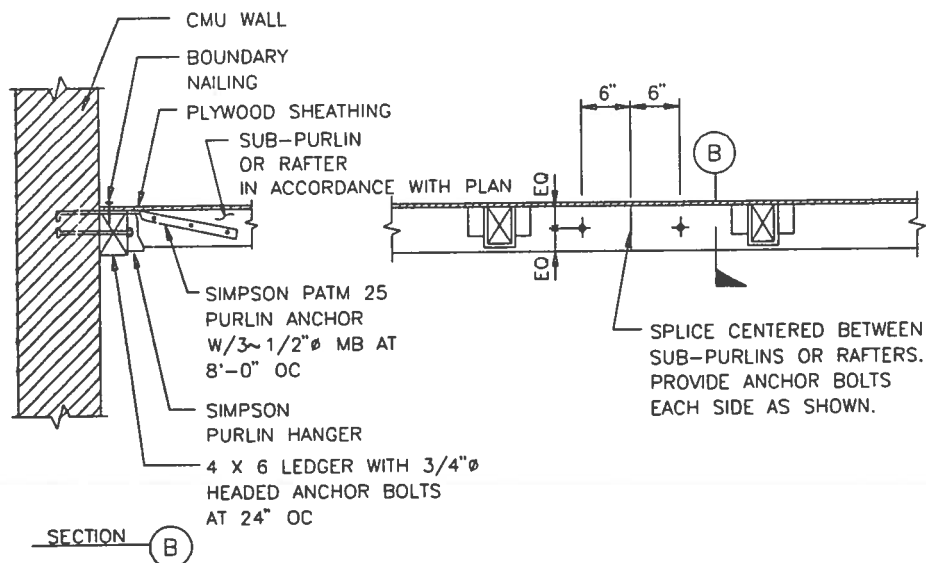


Figure 2: Wood Ledger at Sub-Purlins or Rafters



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TYPICAL STEEL LEDGER ANGLE DETAILS

Figure 1. Ledger Elevation at Purlins

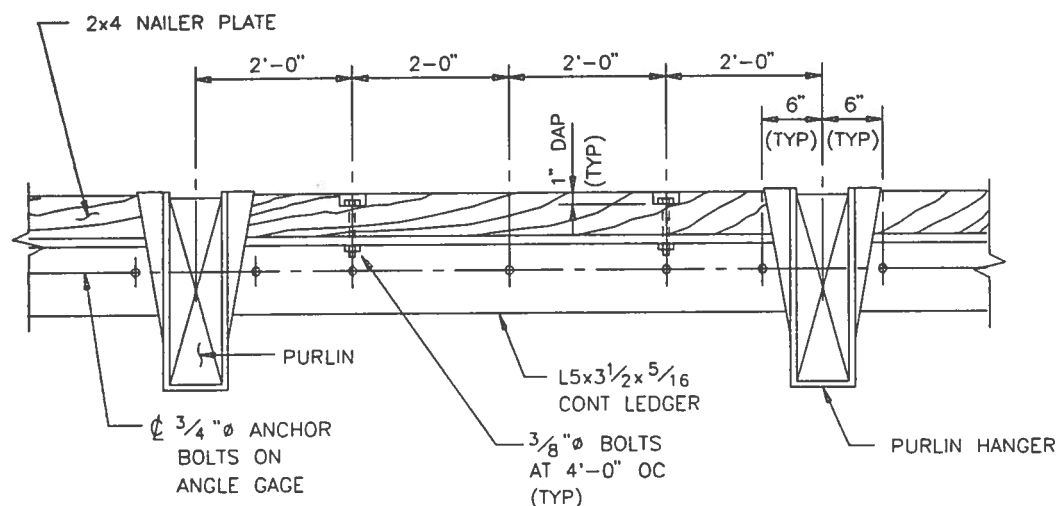


Figure 2. Ledger Elevation at Purlins

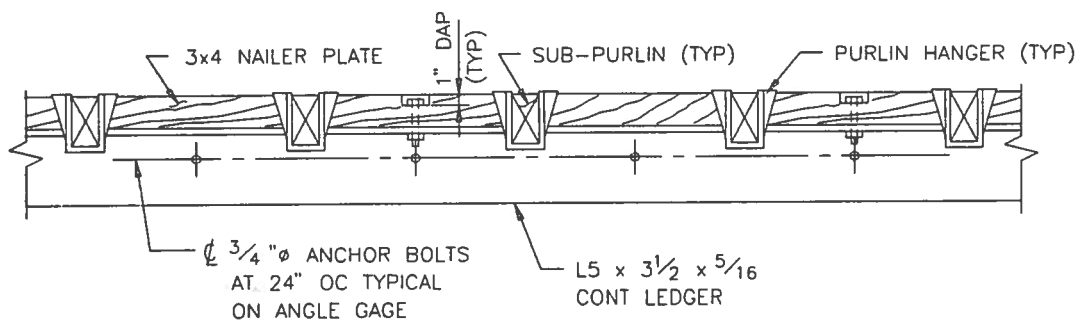
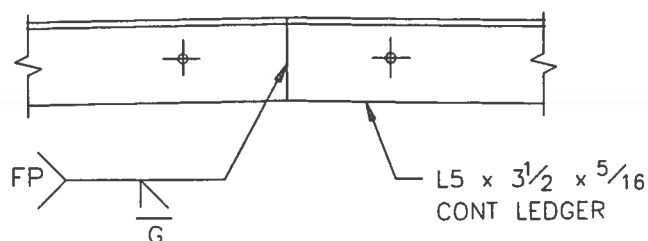


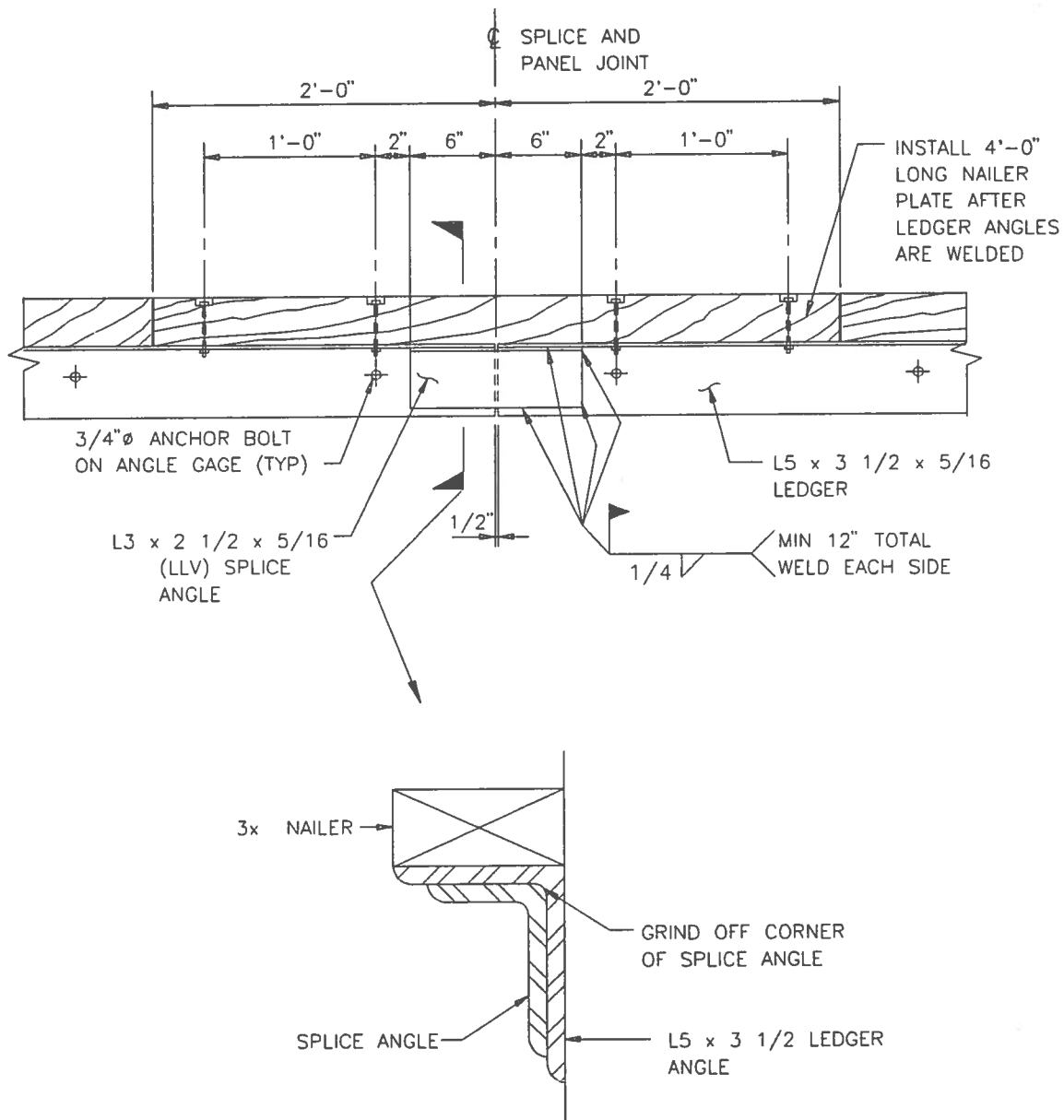
Figure 3. Ledger at Change in Roof Slope



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TYPICAL STEEL LEDGER ANGLE DETAILS

Figure 4. Splice at Ledger Angle



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FIGURE 1: TYPICAL WOOD NAILER
FIGURE 2: TYPICAL BEARING PLATE

Figure 1

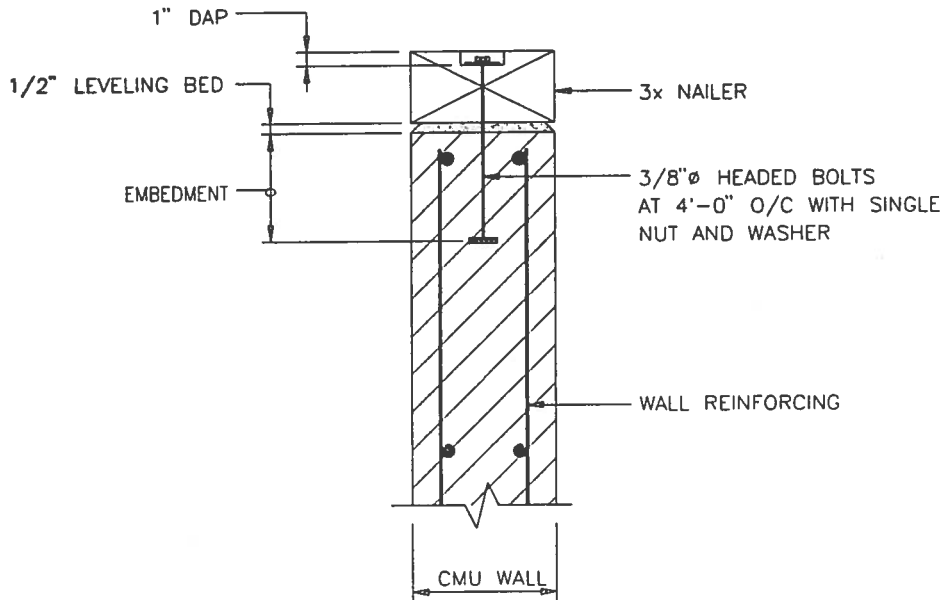
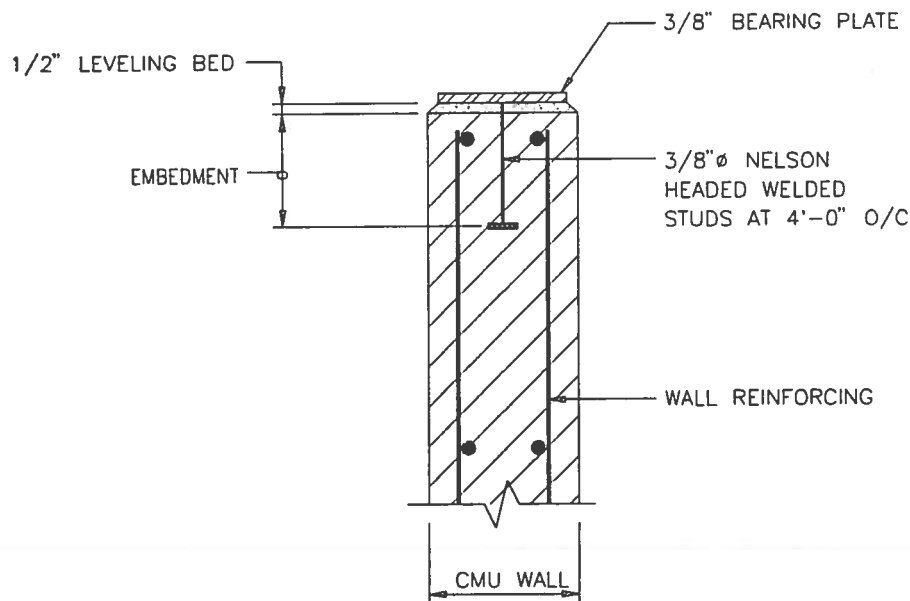


Figure 2



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TYPICAL WOOD FRAMING DETAILS

Figure 1. Wood Framing Perpendicular to Single Story Wall #1

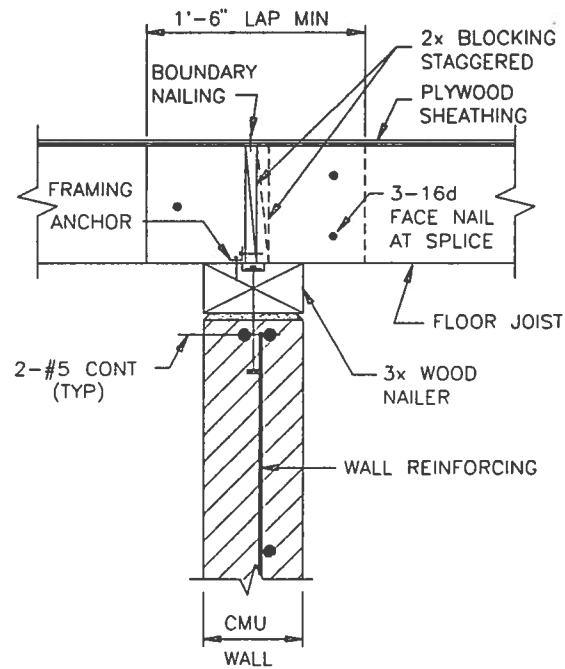
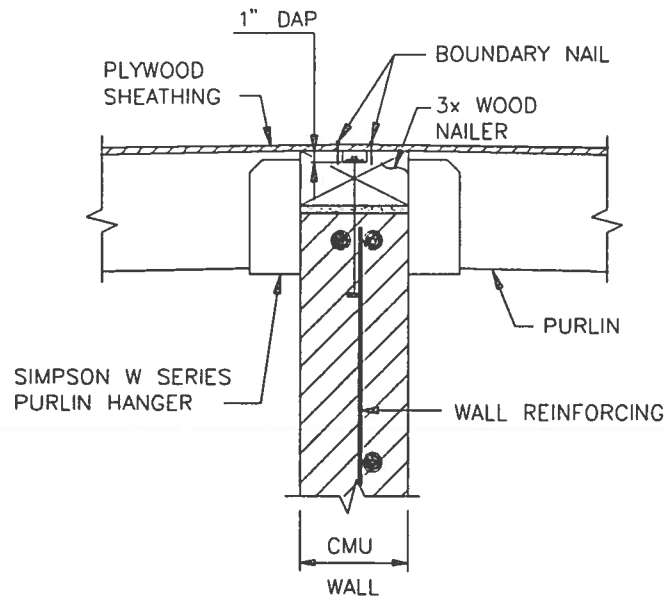


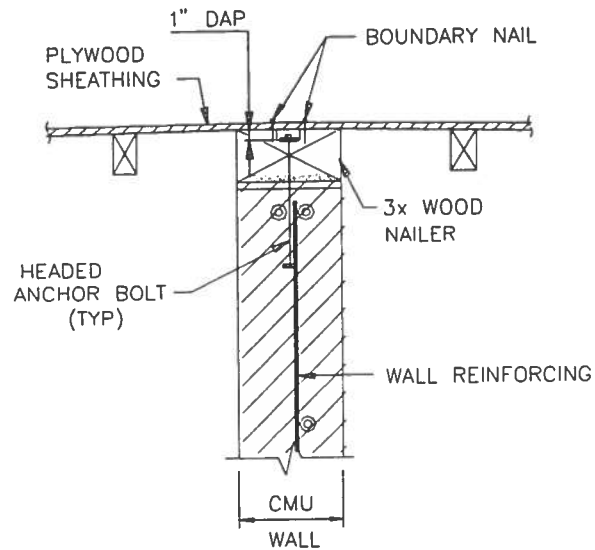
Figure 2. Wood Framing Perpendicular to Single Story Wall #2



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TYPICAL WOOD FRAMING DETAILS

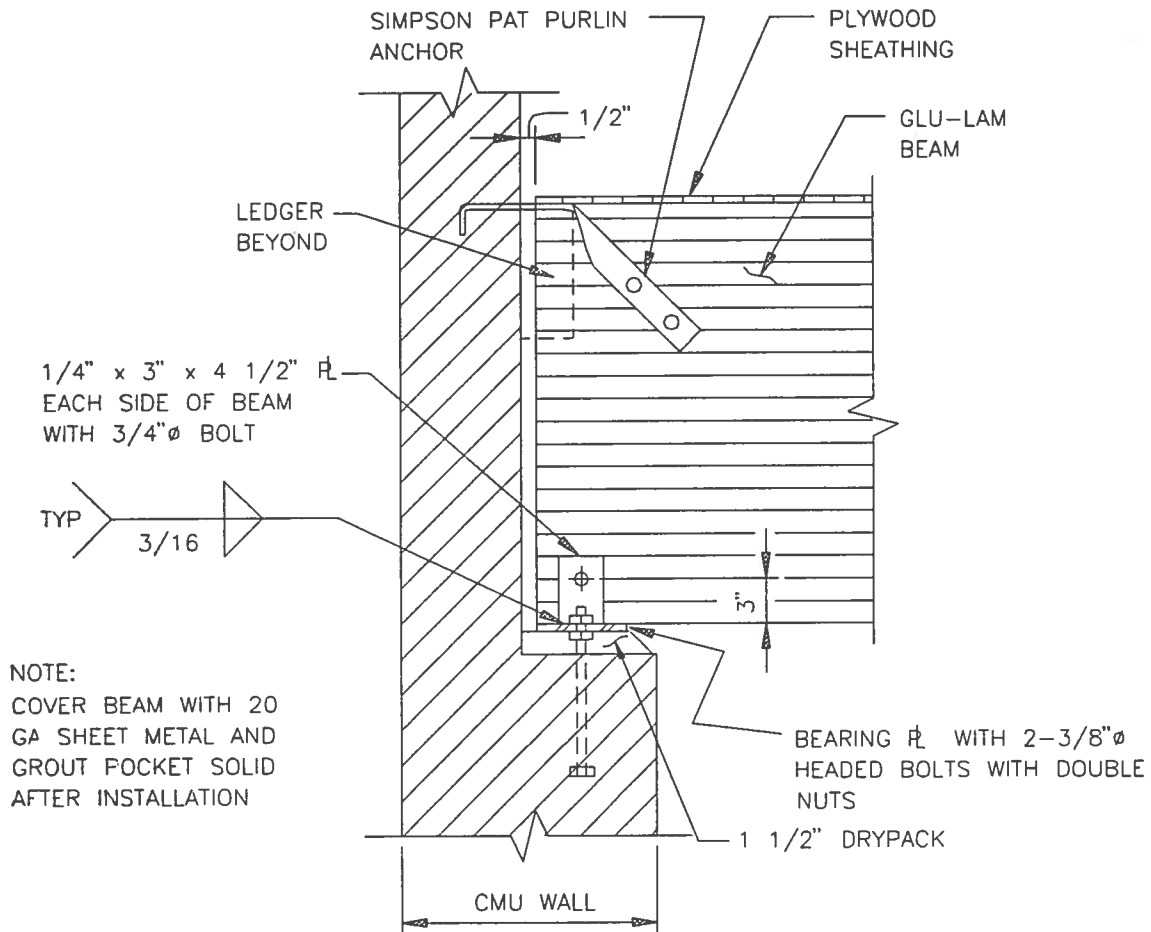
Figure 3. Wood Framing Parallel to Single Story Wall



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FIGURE 1: GLB BEAM SEAT
FIGURE 2: STEEL BEAM SEAT

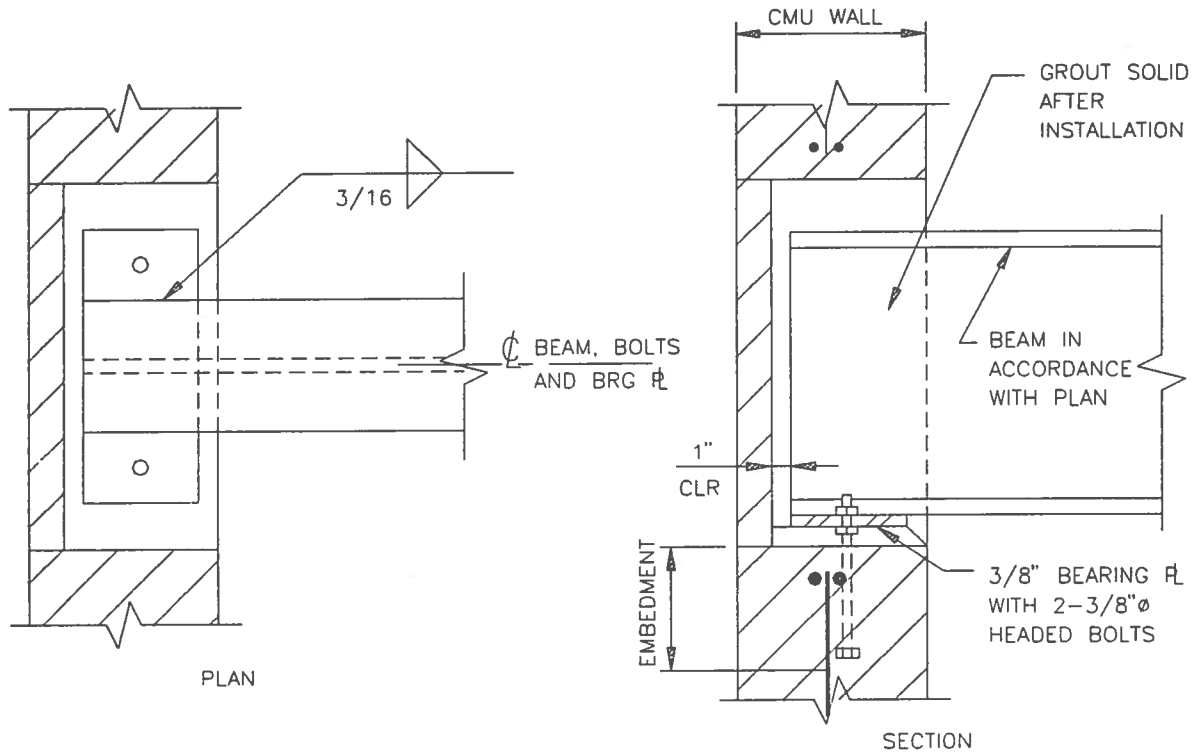
Figure 1



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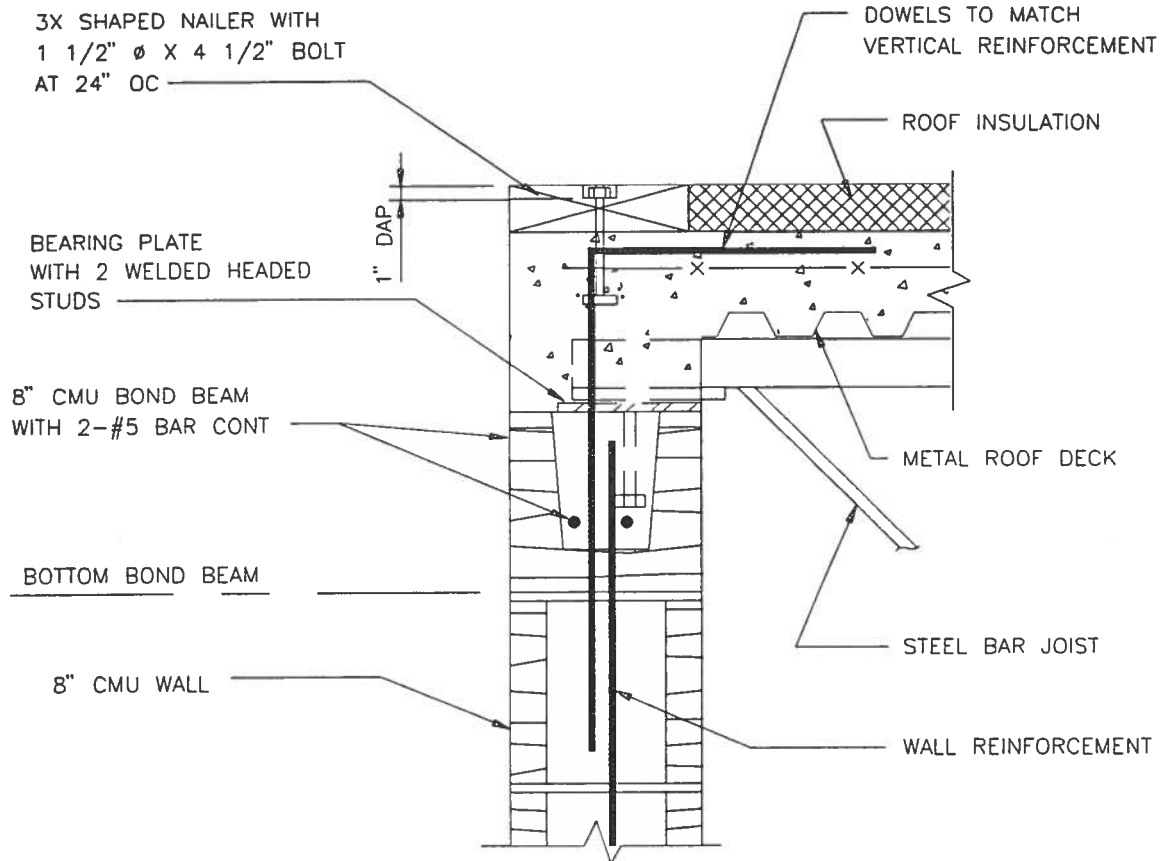
FIGURE 1: GLB BEAM SEAT
FIGURE 2: STEEL BEAM SEAT

Figure 2



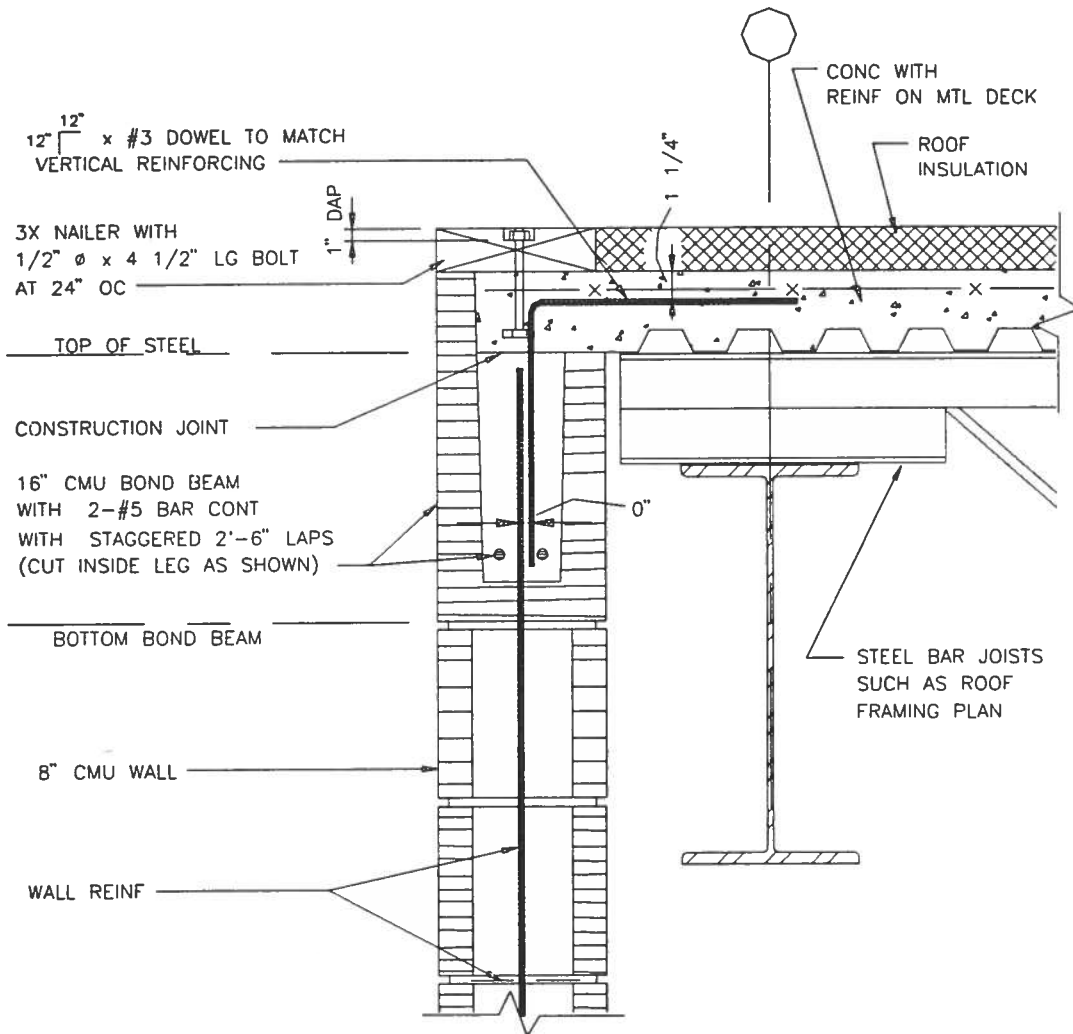
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STEEL JOIST SEAT DETAIL



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STEEL JOIST TO CMU WALL DETAIL

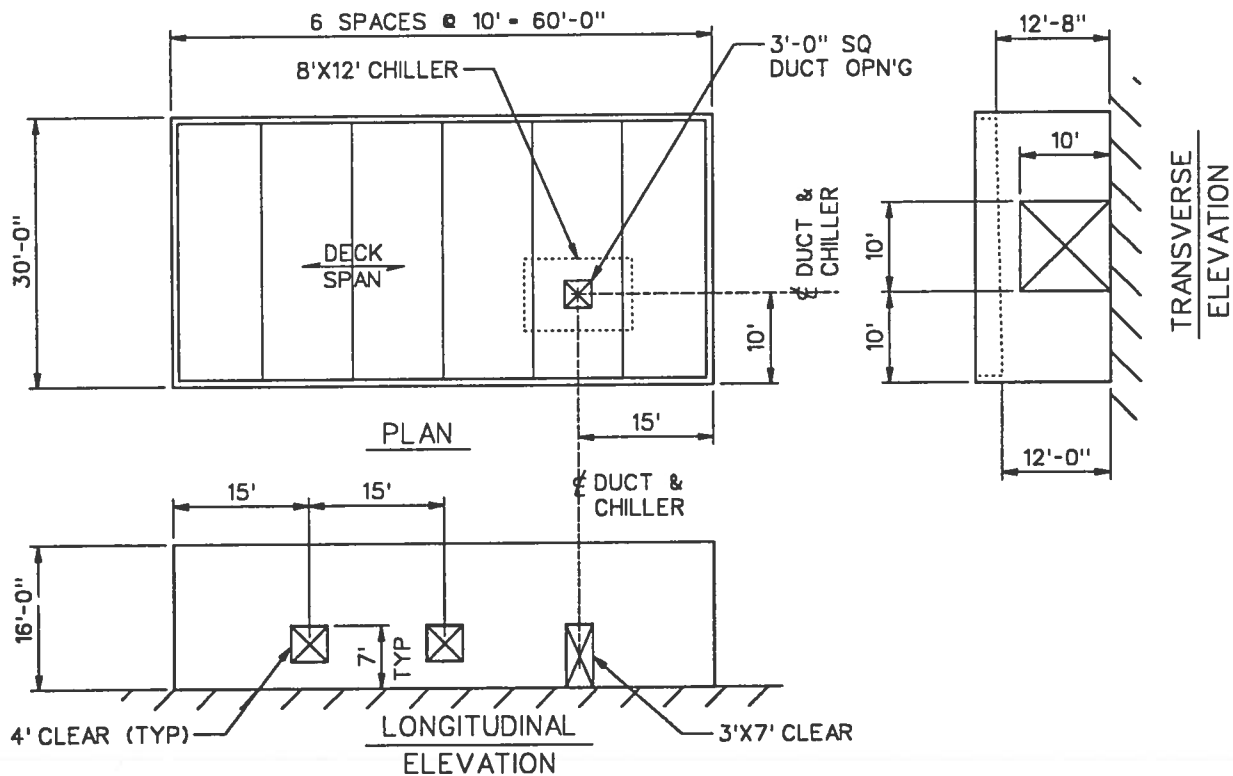


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SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

PROJECT DESCRIPTION

The project consists of a single story reinforced masonry building as shown. The roof consists of metal roof decking supported by steel beams. Beams span transversely to masonry walls. Walls distribute roof loads to continuous wall footings. Lateral resistance is provided by masonry shear walls. The roof deck is analyzed as a flexible diaphragm, distributing loads by tributary areas or statics.



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SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

DESIGN CRITERIA

Code

Uniform Building Code - 1994 Edition
AISC - 8th Edition (ASD)
ACI - 318-95

Lateral

Seismic Zone 4, Given $V = 0.183 W$
Wind Speed = 70 MPH, Given $P = 15.5 \text{ PSF}$

Materials

Concrete: $f'_c = 4000 \text{ PSI}$
Reinforcing: $f_y = 60000 \text{ PSI}$
Steel: A36
Masonry: $f'_m = 1500 \text{ PSI}$ (special inspection required)
Mortar: Type M
Soil Bearing: $SB_{\text{allow}} = 3000 \text{ PSF}$ (1/3 increase allowed for wind and seismic)
 $\mu = 0.35$

Location

Non-Snow region

Roof Unit Loads

Dead

Roofing	= 6.0 PSF
Insulation	= 2.0 PSF
Decking	= 2.5 PSF
Steel Framing	= 2.5 PSF
Misc.	= 1.0 PSF
	= 14.0 PSF

Live

= 20 PSF (reducible)

Roof Deck Design

Span = 10 FT
 $W_{\text{total}} = 6 + 2 + 2.5 + 1 = 11.5 \text{ PSF}$
 $W_{\text{allow}} = 36 \text{ PSF}$
Verify gage and welding with diaphragm analysis.

Use 1-1/2" Roof Deck, Verco HSB - 36, 20 Ga.

FLUOR DANIEL

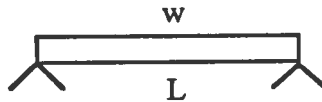
SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Design Roof Beam

w/out chiller

$$L = 30' - 8" = 29' - 4"$$

Tributary width = 10'



$$TA = 10' (29.33') = 293 \text{ FT}^2$$

$$w = 10' (14 + 16) = 300 \text{ PLF} \quad (LL=16 \text{ PSF})$$

$$M = \frac{300 (29.33)^2}{8} = 32.3 \text{ KFT}$$

$$R = \frac{300(29.3)}{2} = 4.4 \text{ K}$$

$$S_{reqd} = 16.2 \text{ in}^3 \quad (\text{try W14X22, } S = 29.0 \text{ in}^3)$$

$$\Delta = \frac{22.5 (300) (29.33)^4}{29 \times 10^6 (199)} = 0.87" = L/415$$

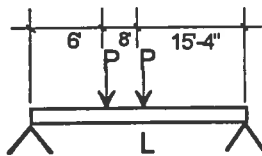
Use W14X22

w/ chiller

$$TA = 293 - 5 (4) = 273 \text{ FT}^2 \quad (LL = 16 \text{ PSI})$$

$$w = 300 \text{ PLF}$$

$$P = 4 \text{ K} / 4 = 1 \text{ K}$$



$$R_1 = \frac{1}{29.33} \left[\frac{0.3(29.33)^2}{2} + (15.33 + 23.33) \right] = 5718 \text{ K}$$

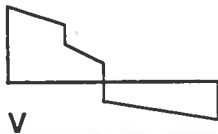
$$R_2 = 0.3(29.33) + 2(1 \text{ K}) - 5.72 = 5.08 \text{ K}$$

$$X = 5.08 / 0.3 = 17' > 15.33$$

$$M = 15.33(5.08) - 0.3(15.33)^2 / 2 = 42.6 \text{ KFT}$$

$$(S_{reqd} = 21.3 < 29.0) \quad (\text{Try W14x22})$$

$$w_{eq} = 42.6(8) / 29.33^2 = 400 \text{ PLF (approx)}$$



FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Design Roof Beam (Continued):

$$\Delta = \frac{0.4(0.87")}{0.3} = 1.16" = L/350$$

Use W14X22

Design Bearing Plate

$$P_{max} = 5720 \text{ LBS}$$

$$F_{br} = 0.38 f_m = 0.38(1500) = 570 \text{ PSI}$$

$$A_{reqd} = (5720)/570 = 10 \text{ in}^2$$

$$b_{min} = 2(1.5") = 3.0"$$

$$L_{min} = 3(2) + 5 = 11"$$

$$A_{prov} = 3(11) = 33 \text{ in}^2 > A_{reqd} \quad (\text{OK})$$

Design plate to develop full capacity of W14x22. (S=29)

$$M_{all} = 58 \text{ KFT}$$

$$w_{max} = [58(8)] / 29.33 = 540 \text{ PLF}$$

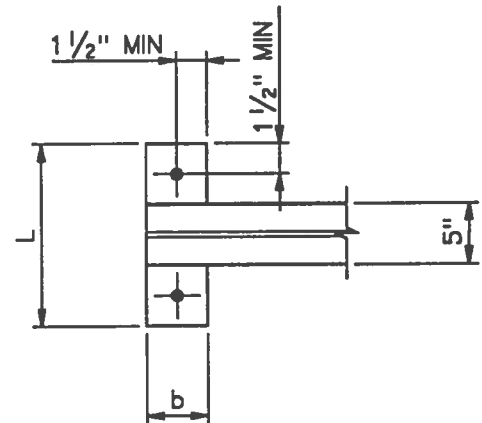
$$R = 540(29.33) / 2 = 7920 \text{ LBS}$$

$$f_{br} = 7920 / 33 = 240 \text{ PSI}$$

$$M = 240(3)^2 / 2 = 1080 \text{ LB-IN/IN}$$

$$t = \sqrt{\frac{6(1080)}{1"(29000)}} = 0.49 \text{ (say } 1/2")$$

Use PL 1/2" X 3" X 0'-11"



Wall Design

Given:

$$F_p = 0.3 W_p \text{ (seismic)}$$

$$8" \text{ Solid grout} = 84 \text{ PSF}$$

$$w_p = 0.3(34) = 25.2 \text{ PSF}$$

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Wall Perpendicular to Deck

$$h = 12.67'$$

$$h/t = [12.67(12)] / 7.625 = 19.9 \quad (\text{OK})$$

$$M = [25.2(12.67)^2] / 8 = 506 \text{ LB-FT}$$

$$P_{\text{midheight}} = 10/2(14) + \left[\frac{12.67}{2} + 4 \right](84) = 940 \text{ LB}$$

$$1.33 K = 506(12)/12(3.75)^2 = 36$$

From table #6, $\rho = 0.0013$

$$A_s = 0.0013(12)(3.75) = 0.059 \text{ in}^2/\text{ft}$$

$$A_{s,\text{min}} = 0.0007(12)(7.625) = 0.064 \text{ in}^2/\text{ft} \quad \leftarrow \text{Governs}$$

(#4 @ 18" c/c, $A_s = 0.133 \text{ in}^2/\text{ft}$)

$$\rho = 0.133/(12(3.75)) = 0.00296 \quad j = 0.903, 2/jk = 7.64$$

$$f_b = \frac{506(12)}{12(3.75)^2}(7.64) = 275 \text{ PSI}$$

$$F_b = 0.33 f_m = 0.33(1500) = 495 \text{ PSI} > f_b \quad (\text{OK})$$

$$f_s = \frac{506(12)}{0.133(0.903)3.75} = 13483 \text{ PSI}$$

$$F_s = 0.5 F_y = 0.5(60) = 30000 \text{ PSI} > f_s \quad (\text{OK})$$

(Use 24,000 PSI)

$$f_a = 940/[12(7.625)] = 10.3 \text{ PSI}$$

$$F_a = 0.2(1500) \left[1 - \left(\frac{12.67(12)}{42(7.625)} \right)^3 \right] = 268 \text{ PSI} > f_a \quad (\text{OK})$$

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} = 10.3/268 + 275/495 = 0.6 < 1.33 \quad (\text{OK})$$

Horizontal Reinforcing,

$$A_s = 0.002 - 0.133/(12(7.625)) = 0.0005 \text{ in}^2/\text{ft}$$

$$= 0.064 \text{ in}^2/\text{ft} \quad \leftarrow \text{Governs}$$

Use # 4 @ 24", $A_s = 0.1 \text{ in}^2/\text{ft}$

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Check Parapet

$$F_p = 2(0.3)W_p/0.75 = 0.8 W_p$$

$$w = 0.8(84) = 67.2 \text{ PSF}$$

$$M = 67.2(4)^2/2 = 537.6 \text{ FT-LB} \quad > 506 \text{ FT-LB}$$

$$P = 4(84) = 340 \text{ LB}$$

$$f_b = \frac{540(12)}{12(3.75)^2} (7.64) = 294 \text{ PSI} < F_b \quad (\text{OK})$$

$$f_a = 340/[12(7.625)] = 4 \text{ PSI}$$

$$h' = 2(4') = 8' < 12.67' \quad \text{therefore, } F_a > 268 \text{ PSI}$$

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} = 4/268 + 294/495 = 0.61 < 1.33 \quad (\text{OK})$$

Use 8" CMU w/ #4 @ 18" c/c vertically, #4 @ 24" c/c horizontally.

Wall Parallel to Deck

$$P_{\text{beam}} = 7920 \text{ LB}$$

$$\begin{aligned} \text{Effective Column} &= \text{Brg. Length} + 4t \\ &= 11" + 4(7.625) \\ &= 41.5" = 3.45' \end{aligned}$$

$$A_g = 7.625 (41.5) = 316.44 \text{ in}^2$$

$$P_{\text{midheight}} = 7920 + 84(3.45)(12.67/2 + 4) = 10915 \text{ LBS}$$

$$f_a = 10915/316.44 = 34.5 \text{ PSI} < F_a = 268 \text{ PSI} \quad (\text{OK})$$

$$f_b = 294 \text{ PSI} < F_b = 495 \text{ PSI} \quad (\text{OK})$$

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} = 34.5/268 + 294/495 = 0.73 < 1.33 \quad (\text{OK})$$

Use 8" CMU Wall w/ #4 @ 18" c/c vertically, #4 @ 24" c/c horizontally.

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Lintel Beam

10'-0" Span

$$w = (10/2) (14 + 20) + 6(85) \\ = 680 \text{ PLF}$$

$$M = [680 (10)^2] / 8 = 8500 \text{ LB-FT}$$

Check 2 - #5 ($A_s = 0.62 \text{ in}^2$)

$$\rho = 2 (0.31) / [7.625 (66)] = 0.0012$$

$$d = 6'(12) - 6" = 66"$$

From table #7,

$$j = 0.911, 2/jk = 8.19 \text{ (table \#6)}$$

$$f_b = \frac{8500(12)}{7.625(66)^2} (8.19) = 25.2 \text{ PSI} < F_b = 0.33 f_m = 495 \text{ PSI} \quad (\text{OK})$$

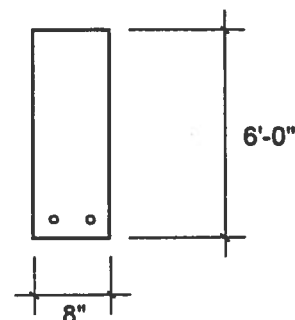
$$f_s = \frac{8500(12)}{0.62(0.911)66} = 2740 \text{ PSI} < F_s = 24000 \text{ PSI} \quad (\text{OK})$$

$$V = 680(10')/2 = 3400 \text{ LBS}$$

$$f_v = \frac{3400}{7.625(0.911)(66)} = 7.4 \text{ PSI} < F_v = \sqrt{1500} = 38.7 \text{ PSI} \quad (\text{OK})$$

Therefore, no shear rebar is required.

Use 2 - #5



Lateral

Seismic (Given $V = 0.183 W$)

A. Longitudinal

Masses:

Roof (14 PSF)(30)(60)	= 25.2 K
Chiller	= 8.0 K
Walls (84 PSF)(12/2 + 4)(30)2	= 50.4 K
	= 83.6 K

$$V_L = 0.183 (83.6) = 15.4 \text{ K}$$

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Seismic

B. Transverse

Masses:

Roof	= 25.2 K
Chiller	= 8.0 K
Walls (50.4)(60)/30	= 100.8 K
	= 134 K

$$V_T = 0.183 (134) = 25.6 \text{ K}$$

Wind

(Given P = 15.5 PSF)

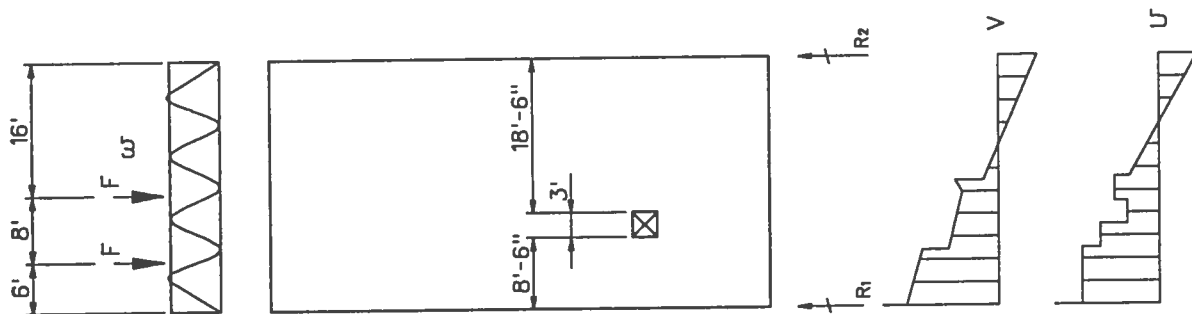
A. Longitudinal

$$P_L = 15.5 (30') (16') = 7.2 \text{ K} < 15.4 \text{ K} \quad (\text{Seismic Governs})$$

B. Transverse

$$P_T = 7.2 \text{ K} (2) = 14.4 \text{ K} < 25.6 \text{ K} \quad (\text{Seismic Governs})$$

Diaphragm



A. Longitudinal

$$w = 0.183 (25.2 + 50.4)/30' = 462 \text{ PLF}$$

$$F = 0.183 (8)/2 = 733 \text{ LBS}$$

$$R_1 = \frac{1}{30} \left[\frac{462(30)^2}{2} + 733(16 + 24) \right] = 7910 \text{ LBS}$$

$$R_2 = \frac{1}{30} \left[\frac{462(30)^2}{2} + 733(6 + 14) \right] = 7415 \text{ LBS}$$

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Diaphragm (Continued):

Note shape of shear diagram V vs. Unit Shear Diagram v. For simplicity, calculate v_{max} from maximum shear and reduced length.

$$v_{max} = \frac{7910}{60' - 2(\frac{8}{12}) - 3'} = 143 \text{ PLF}$$

B. Transverse

$$w = 0.183 (25.2 + 100.8)/60' = 385 \text{ PLF}$$

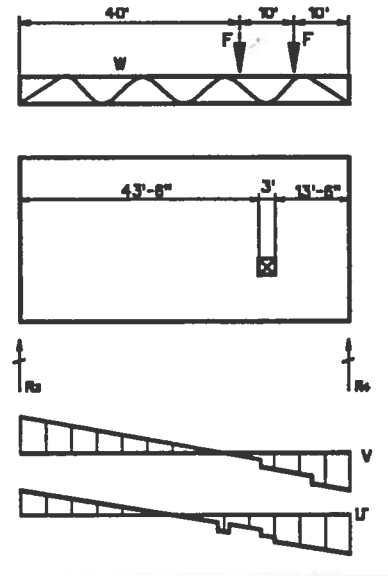
$$F = 733 \text{ LBS}$$

$$R_3 = \frac{1}{60} \left[\frac{385 (60)^2}{2} + 733(10 + 20) \right] = 11915 \text{ LBS}$$

$$R_4 = \frac{1}{60} \left[\frac{385 (60)^2}{2} + 733(40 + 50) \right] = 12650 \text{ LBS}$$

$$V_{max} = 12650/(30 - 1.33 - 3) = 493 \text{ PLF (perpendicular)}$$

$$V_{allow} = 750 \text{ PLF}$$



Use HSB - 36 Deck, 18 Ga. 5 - 3/4" welds per unit. Top seam weld 1-1/2" @ 24" c/c.

Shearwalls

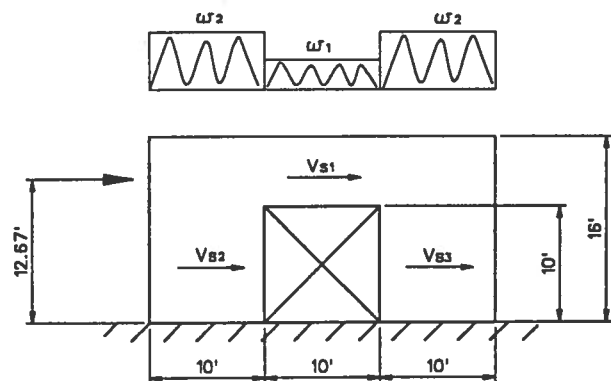
N-S Direction

$$V = 12.65 \text{ K}$$

$$V_{s1} = 0.183(6')(30')(85 \text{ PSF}) = 2.8 \text{ K}$$

$$V_{s2, s3} = 0.183(10')^2 (85) = 1.56 \text{ K}$$

Piers are of equal size and rigidity, therefore, loads are equally shared.



$$V_{pier} = 12.65 \text{ K} + 2.8/2 + 1.56 \text{ K} = 15.61 \text{ K}$$

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Overturning (gross)

$$M_{OT} = 12.65(12.67') + 2.8(13') + 2(1.56)(5') \\ = 212.3 \text{ KFT}$$

$$w_1 = 5'(14) + 6'(85) = 580 \text{ PLF}$$

$$w_2 = 70 \text{ PLF} + 16'(85) = 1360 \text{ PLF} \quad \Delta = 780 \text{ PLF}$$

$$M_R = \frac{580(30)^2}{1000(2)} + \frac{780(10)(5'+25')}{1000} = 495 > M_{OT}$$

$$SR = 495/212 = 2.33 \quad (\text{OK})$$

Check Uplift

Try 2 - #5

$$d = 10'(12) - 3" = 117"$$

$$\rho = 0.31 (2) / [7.625 (117)] = 0.0007$$

$$\rho_n = 0.0139$$

$$\text{Table \#8a} \quad j = 0.949 \quad 2/jk = 13.70$$

$$M = M_u/2 = 106150 \text{ KFT}$$

$$f_b = \frac{106150(12)}{7.625(117)^2} (13.7) = 167 \text{ PSI}$$

$$< F_b = 495 \text{ PSI} \quad (\text{OK})$$

$$f_s = 106150(12)/[0.62(0.949)(117)] = 18503 \text{ PSI}$$

$$< F_s = 20000 \text{ PSI} \quad (\text{OK})$$

$$\frac{M}{Vd} = 106150 (12)/[15610(117)] = 0.69$$

$$< 1.0$$

$$f_v = \frac{1.5(15610)}{7.625(0.949)(117)} = 27.7 \text{ PSI}$$

$$F_v = \frac{1}{2}(4 - 0.69)\sqrt{1500} = 64 \text{ PSI} \\ = 120 - 45(0.69) = 89 \text{ PSI}$$

← Governs

$$> f_v \quad (\text{OK})$$

No shear reinforcing is required.

Use 2 - #5 Jamb bars

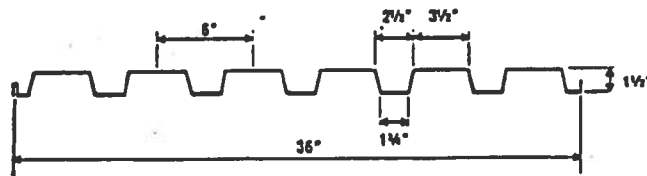
FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Type HSB-36

High Shear B Deck-36" Wide
Verco's New Standard B Deck
Higher lateral load capacity
Also available 24" or 30" wide with button punch or overlapping side lap on special order

FACTORY MUTUAL APPROVED
See page 14, Item 14



Gage	Weight (Lbs./Sq. Ft.)		I (in ⁴)	+ S (in ³)	- S (in ³)
	Prime Painted	Galvanized			
22	1.8	1.9	.183	.209	.209
20	2.2	2.3	.233	.271	.271
18	2.8	2.9	.338	.395	.395
16	3.4	3.5	.440	.502	.502

VERTICAL LOADS (Lbs./Sq. Ft.) HSB-36 AND HSB-36 WITH SHEARTRANZ®

PRIME PAINTED OR GALVANIZED

	Gage		Span												
			5'-0"	5'-6"	6'-0"	6'-6"	6'-8"	7'-0"	7'-6"	8'-0"	8'-4"	8'-6"	9'-0"	9'-6"	10'-0"
Simple Span	22	s	105	86	73	62	59	53	46	41	38	36			
		Δ	95	71	55	43	40	34	28	23	20	19			
	20	s	131	108	91	78	74	67	58	51	47	45	40		
		Δ	115	88	68	53	49	43	35	28	25	24	20		
	18	s	179	148	124	106	101	91	80	70	65	62	55	50	45
		Δ	164	123	95	75	70	60	48	40	35	33	28	23	20
	16	s	227	188	158	134	128	118	101	88	81	78	70	63	56
		Δ	204	153	118	93	89	74	60	50	44	41	35	30	25
Two Spans	22	s	110	91	77	65	62	56	49	43	40	38	34		
		Δ	—	—	—	—	—	—	—	—	—	—	—		
	20	s	140	116	97	83	79	71	62	55	50	48	43	39	36
		Δ	—	—	—	—	—	—	—	—	—	—	—	—	36
	18	s	185	153	128	109	104	94	82	72	67	64	57	51	46
		Δ	—	—	—	—	—	—	—	—	—	—	—	—	—
	16	s	230	190	160	136	129	117	102	90	83	78	71	63	57
		Δ	—	—	—	—	—	—	—	—	—	—	—	—	—
Three or More Spans	22	s	138	114	96	82	78	70	61	54	50	48	43		
		Δ	—	—	—	81	75	65	53	43	38	36	30		
	20	s	176	145	122	104	99	89	78	68	63	61	54	48	45
		Δ	—	—	—	101	94	81	65	54	48	45	38	32	25
	18	s	232	191	161	132	127	118	103	90	83	80	71	64	58
		Δ	—	—	—	—	—	112	91	75	66	62	52	45	38
	16	s	288	238	200	170	162	146	128	112	103	99	89	80	72
		Δ	—	—	—	—	162	140	113	94	83	78	65	56	48

See page 14, Item 1 on how to use this table.

FLUOR DANIEL

SAMPLE DESIGN 1: MASONRY BUILDING DESIGN

Type HSB-36

DIAPHRAGM SHEAR VALUES (q) (Lbs./L.F.) AND FLEXIBILITY FACTORS (F)

PRIME PAINTED OR GALVANIZED

Sage	Side Lap Connection	Welds Per Span To Support	Span											
			5'-0"	6'-0"	7'-0"	8'-0"	9'-0"	10'-0"						
22	BP @ 24" c/c	Q	260	430	240	400	240	380	230	370	230	340		
		F	5.5+	5.0+	6.9+	6.1+	8.4+	7.5+	10.3+	8.3+	12.4+	11.4+		
	BP @ 12" c/c	Q	310	480	300	460	300	440	290	430	290	400		
		F	5.0+	4.7+	6.0+	5.7+	7.3+	6.9+	8.7+	8.4+	10.4+	10.1+		
	TSW @ 24" c/c	Q	190	320	130	230	120	200	120	190	110	190		
		F	5.6+	5.3+	6.3+	5.8+	6.5+	6.2+	6.6+	6.5+	6.6+	6.8+		
	TSW @ 12" c/c	Q	800	890	800	860	800	870	700	860	800	740		
		F	5.2+	5.1+	5.3+	5.2+	5.3+	5.4+	5.2+	5.4+	5.1+	5.4+		
20	BP @ 24" c/c	Q	390	670	360	610	360	530	350	470	320	420		
		F	5.7+	4.5+	6.6+	5.7+	8.2+	7.2+	10.2+	9.0+	12.5+	11.2+		
	BP @ 12" c/c	Q	450	730	420	690	430	600	420	530	340	440		
		F	5.3+	4.8+	5.9+	5.4+	7.2+	6.7+	8.8+	8.3+	10.7+	10.1+		
	TSW @ 24" c/c	Q	620	790	540	640	520	630	520	610	540	610		
		F	5.4+	4.9+	5.9+	5.4+	6.2+	5.9+	6.3+	6.2+	6.3+	6.4+		
	TSW @ 12" c/c	Q	780	890	750	840	780	840	620	860	660	800		
		F	4.9+	4.6+	4.9+	4.8+	4.9+	5.0+	4.9+	5.1+	4.8+	5.1+		
18	BP @ 24" c/c	Q	750	1050	640	800	550	740	490	850	440	590	400	530
		F	5.3+	4.5+	6.2+	5.3+	7.9+	6.8+	10.0+	8.7+	12.5+	10.9+	15.3+	13.6+
	BP @ 12" c/c	Q	830	1110	720	930	630	800	560	710	500	640	450	580
		F	5.0+	4.4+	5.7+	5.0+	7.2+	6.4+	9.0+	8.1+	11.1+	10.2+	13.6+	12.6+
	TSW @ 24" c/c	Q	890	1120	760	950	640	840	570	720	510	650	460	590
		F	4.9+	4.3+	5.5+	4.9+	5.8+	5.4+	6.0+	5.8+	6.0+	6.1+	6.0+	6.3+
	SSW @ 24" c/c	Q	1020	1370	1010	1200	920	1080	640	990	810	930	770	870
		F	6.7+	6.1+	6.1+	5.3+	5.9+	5.2+	5.7+	5.1+	5.5+	4.9+	5.4+	4.8+
	TSW @ 12" c/c	Q	1050	1250	1010	1140	1040	1140	1080	1150	1160	1190	1250	1280
		F	4.4+	4.1+	4.5+	4.4+	4.5+	4.6+	4.7+	4.4+	4.7+	4.3+	4.7+	4.3+
	SSW @ 12" c/c	Q	1570	1810	1470	1690	1390	1550	1330	1460	1280	1390	1240	1340
		F	5.4+	5.1+	4.7+	4.5+	4.5+	4.4+	4.3+	4.2+	4.2+	4.1+	4.0+	4.0+
16	BP @ 24" c/c	Q	1000	1350	820	1110	700	950	620	840	550	750	500	680
		F	5.4+	4.5+	6.0+	5.0+	7.8+	6.5+	9.9+	8.5+	12.5+	10.8+	15.5+	13.5+
	BP @ 12" c/c	Q	1080	1300	900	1170	780	1010	640	890	610	800	590	720
		F	5.1+	4.4+	5.5+	4.8+	7.2+	6.2+	9.1+	8.0+	11.4+	10.2+	14.1+	12.8+
	TSW @ 24" c/c	Q	1130	1340	920	1200	820	1040	660	900	610	800	590	720
		F	5.0+	4.4+	5.2+	4.6+	5.6+	5.1+	5.8+	5.5+	5.9+	5.9+	6.0+	6.1+
	SSW @ 24" c/c	Q	1480	1890	1260	1560	1010	1310	770	1140	740	940	730	940
		F	4.3+	4.4+	4.3+	4.0+	4.2+	3.9+	4.0+	3.8+	3.9+	3.8+	3.7+	3.7+
	TSW @ 12" c/c	Q	1330	1590	1260	1460	1010	1240	770	1040	740	940	730	940
		F	4.5+	4.1+	4.3+	4.1+	4.4+	4.3+	4.3+	4.3+	4.3+	4.2+	4.2+	4.6+
	SSW @ 12" c/c	Q	2150	2650	2020	2410	1530	2020	1240	1810	1210	1710	1210	1710
		F	3.8+	3.7+	3.6+	3.4+	3.5+	3.3+	3.3+	3.2+	3.2+	3.2+	3.1+	3.1+

WELD PATTERN TO SUPPORTS

